

DreamRole Project Audit

Comprehensive technical breakdown and current implementation status

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****Comprehensive Project Audit & Technical Specification****

1. Project Mission

DreamRole is designed to bridge the chasm between current workforce capabilities and evolving industry requirements. By leveraging Large Language Models (LLMs) and a curated dataset of 320+ industry roles, the platform provides personalized, actionable insights to help users achieve "Role Ready" status.

2. Core Features & Capabilities



Intelligence Resume Parsing

- **PDF Extraction****: Converts complex PDF layouts into clean text using ``pdf-parse``.
- **AI Skill Extraction****: Uses OpenAI GPT models to identify technical and soft skills from raw resume data, mapping them to industry-standard taxonomies.



Precision Skill Gap Analysis

- **Domain-Specific Logic****: Users select from 30+ domains and 320+ specific roles.
- **Core Skill Matching****: Compares user skills against a curated dataset where each role is defined by 6-8 high-impact core skills.
- **Alignment Stages****: Categorizes users into four progress levels:
 - **Foundation Stage (0-29%)****: Basic understanding, needs significant building.
 - **Developing Stage (30-54%)****: Intermediate knowledge, missing several core pillars.
 - **Skilled Stage (55-79%)****: Strong alignment, ready for advanced projects.
 - **Role Ready Stage (80-100%)****: Industry alignment achieved.

Adaptive AI Evaluation

- **Dynamic MCQs**: Generates role-specific multiple-choice questions on-the-fly to validate claimed skills.
- **Knowledge Verification**: Questions are weighted based on the complexity of the "missing skills" identified during analysis.

Growth Roadmap & Recommendations

- **Actionable Feedback**: AI-generated prose detailing how to improve specific weak areas.
- **Project Recommendations**: Curated list of GitHub-ready projects tailored to the target role.
- **Micro-Certifications**: Linkage to industry-recognized certifications (Coursera, Udemy, etc.).

Mentor Discovery & Community

- **Expert Mapping**: Access to a database of 16+ industry mentors categorized by domain.
- **Progress Tracking**: Historical view of evaluation attempts and skill growth over time.

3. Technical Stack

| Layer | Technology | | :

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| | **Frontend** | React.js (Vite), Tailwind CSS (Rich Aesthetics), Lucide Icons | | **State Management** | React Context API (AppContext) | | **Visualization** | Recharts (for progress and skill charts) | | **Backend** | Node.js, Express.js | | **AI/ML** | OpenAI GPT API (extractSkills, generateTest, generateFeedback) | | **Database** | CSV Data Layer (Job Roles), JSON (Progress & Mentors) | | **System Tools** | Puppeteer (PDF Reporting), Multer (Storage), pdf-parse |

4. System Architecture

The application follows a modular **Client-Server** architecture:

- ****Client****: A highly interactive SPA (Single Page Application) that manages a unified "Workflow Stepper" state.
 - ****Server****: A stateless REST API handling heavy computation, AI interfacing, and document generation.
 - ****Service Layer****: Dedicated services for OpenAI integration, PDF generation, and CSV data handling ensure clean separation of concerns.
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5. API Reference (Key Endpoints)

- `POST /api/resume/upload``: Ingests resumes and returns raw text.
 - `POST /api/skills/extract``: AI-driven extraction of skill entities.
 - `POST /api/analysis``: Core engine for gap comparison and stage classification.
 - `POST /api/test/generate``: On-the-fly generation of evaluation questions.
 - `POST /api/report/generate``: Server-side PDF generation using Puppeteer.
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6. Current Implementation Status

- **COMMITTED** Full Frontend Stepper Workflow (UI/UX)
 - **COMMITTED** OpenAI Backend Integration (Logic)
 - **COMMITTED** Skill Gap & Alignment Engine (Algorithm)
 - **COMMITTED** Real-time Progress Tracking (Database)
 - **COMMITTED** CSV Job Role Dataset (320+ Roles)
 - **COMMITTED** Professional PDF Report Export (Service)
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Report generated by Antigravity AI for DreamRole Future Planning.
